# EFFICIENT FORMULATION AND SOLUTION OF NONLINEAR MODEL PREDICTIVE CONTROL PROBLEM

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## Abstract

**Large Scale optimization strategies for process operation have evolved significantly over the past two decades. Currently, continuous variable optimization problems (nonlinear programs) can be solved on-line for steady state refinery models with several hundred thousand variables. Moreover, efficient NLP strategies have been developed for dynamic optimization problems. Still, the next step, on-line optimization of large dynamic chemical processes, requires the tackling of a number of limitations and research challenges. Many of the advances in NLP algorithms have taken place by recognizing and exploiting the framework of Successive Quadratic Programming (SQP) algorithms. These are extensions of Newton type methods for converging to the solution of the KKT (optimality) conditions of the optimization problem. Moreover, a number of innovations in algorithm design and problem formulation can greatly improve performance. As a result, very fast NLP algorithms can be derived for data reconciliation, parameter estimation, nonlinear model predictive control and dynamic optimization. Moreover, inequality constraints and variable bounds can be treated through advances in interior point strategies. These methods preserve the particular problem structure and scale well in performance for large-scale problems with many constraints. In particular, we will consider the application of these strategies to problems in Nonlinear Model Predictive Control (NMPC). In parallel to the development of efficient nonlinear programming algorithms, we also need to consider the incorporation of nonlinear models that are accurate but also easily identified and implemented. Here we consider DABNet (decoupled A-B neural network) models which have very desirable approximation properties for a wide variety of nonlinear systems. These models can be incorporated easily into the NMPC framework and lead to very efficient online control algorithms. Finally, the concepts developed in this paper will be illustrated by several case studies drawn from our previous work. These examples illustrate the importance of nonlinear model predictive control and emphasize the need for efficient online computation.**

## 1. Introduction

Increasing acceptance of model predictive control (MPC) as a general purpose strategy for process control leads us to consider more accurate and complex nonlinear models. Two advantages of MPC are the use of a general control law to handle MIMO systems directly and a framework to handle input and output constraints. Using linear dynamic plant models, MPC has become a standard tool with thousands of commercial applications and successes. Moreover, over the past five years, theoretical properties of MPC have been analyzed more deeply. Based on stability properties of LQG through Lyapunov analysis dating from the early 70s (Kleinman, 1970; Kwon and Pearson, 1977; Keerthi and Gilbert, 1988) the stability properties of MPC are now much better understood. In particular, Rawlings and coworkers (e.g., Rawlings and Muske, 1993; Scokaert *et al.*, 1997; Scokaert and Rawlings, 1997) have identified and analyzed several MPC problem formulations (e.g., infinite horizon, endpoint constrained, dual mode) which inherit these stability properties.

Given the success of MPC with linear models, it is useful to justify the more difficult task of incorporating nonlinear dynamic models for MPC. While MPC with linear models requires the solution of convex problems for each sampling time (usually quadratic programs (QPs) or linear programs (LPs)), nonlinear model predictive control (NMPC) requires the consideration (and at least a partial solution) of a nonconvex, nonlinear program (NLP). Difficulties with NMPC come from both theoretical and computational perspectives. As noted by Mayne (Mayne, 1996), significant concerns are introduced by nonconvexity since nominal stability, which requires strong descent of the Lyapunov function, can no longer be guaranteed. A related concern is the expense and reliability of solving the NLP online.

Despite these concerns, NMPC is being applied increasingly in the chemical industry and these applications are viewed as essential for a number of process control problems. NMPC is needed especially for nonlinear, unsteady batch processes where a trajectory needs to be followed from the prediction of a nonlinear model. It is also needed for processes with wide swings in operation, beyond the ranges of a local linearization. This is especially true for processes operating at or near 'singular' points (e.g., sign changes in process gain and input multiplicities) that cannot be captured by linear controllers and where higher order information is needed.

Model Predictive Control uses the dynamic model to predict the effect of sequences of control steps on the controlled variables. The aim is to derive an optimal set of control steps, which will drive these output variables to desired steady state set points. These set points are based on economic, safety, environmental or product considerations. Moreover, an important consideration for nonlinear model predictive control is its position in the framework of on-line process optimization. Currently, on-line optimization is performed with steady-state nonlinear models and these pass setpoints and feedforward information to controllers represented by linear dynamic models. A more natural approach would be to start with an overall, integrated, dynamic optimization problem. Such a problem could consider the influence of the scheduling and operating policies, the interactions of upstream or downstream processes and the local inputs which could include disturbances and changing performance requirements. Also, at the "lowest" level, the resulting control strategy could still require the solution of nonlinear dynamic optimization problems and it is for the efficient execution of this task, that this paper is directed.

In the following sections, recent advances are covered for the solution of nonlinear dynamic optimization problems which have the characteristics of NMPC problems. This review is by no means complete but we hope to highlight some of the major computational trends and show their relevance to NMPC problem formulations. These become especially important as larger applications and problem formulations are considered.

In the next section we introduce the problem definition and the basic SQP algorithm for NMPC. This is based on a sequential approach which requires the forward solution of the state equations. Section 3 then describes a simultaneous approach for NMPC and discusses a number of refinements that make this strategy very efficient. Further gains are described in Section 4 with the addition of DABNet models. These models have very desirable approximation properties and are described with a linear dynamic model and a nonlinear steady state mapping. A number of NMPC applications that apply DABNet models are presented in Section 5. Finally, section 6 summarizes some of these points and gives directions for future work.

## 2. NMPC Problem Definition and Basic Algorithm

Consider a state space process model of the form:

$$\dot{x} = f(x, z, u), \tag{1}$$

$$0 = c(x, z, u), \tag{2}$$

$$y = g(x), \tag{3}$$

with $f, c$ and $g$ twice continuously differentiable. Here the state vector consists of the differential variables $x \in \mathsf{R}^{\mathsf{n_{sd}}}$, and algebraic variables $z \in \mathsf{R}^{\mathsf{n_{sa}}}$, the input vector is $u \in \mathsf{R}^{\mathsf{n_i}}$ and the output vector is $y \in \mathsf{R}^{\mathsf{n_o}}$. Without loss of generality, assume that the algebraic constraints $c = 0$ can be formulated as index one and, as a result the algebraic equations, $c$, and the algebraic state variables, $z$, can be eliminated. This is done for convenience of notation; direct elimination of algebraic equations is not required as all of the algorithms formulated below can accommodate them implicitly.

For a constant sampling time $T$ it is convenient to define the discrete time operator corresponding to (1, 2) as $x_{i+1} = \chi(T; x_i, u_i)$. This operator can be evaluated explicitly through integration of the DAE model, or approximated and represented implicitly, for instance, using collocation on finite elements. Substitution of the discretized model leads to the following control problem formulation:

$$\begin{aligned} Min_{u_i, x_i, y_i} \quad & \sum_{i=1}^{p} h_i(y_{sp,i} - y_i, u_{i-1} - u_{r,i-1}) \\ s.t.\ & x_i = \chi(x_{i-1}, u_{i-1}), \quad y_i = g(x_i) \quad i = 1, \ldots, p. \\ & u(t) = u_{m-1},\ t \in [t_m, t_p], \quad u_l \le u_i \le u_u \\ & x_l \le x_i \le x_u, \quad y_l \le y_i \le y_u \end{aligned} \tag{4}$$

where the terms in the objective function $h(\xi, \eta)$ are nonnegative, continuously differentiable and take the value zero iff $\xi$ and $\eta$ are zero. A typical example of this objective function is:

$$\begin{aligned} h(\xi, \eta) = & (y_{sp,i} - y_i)^T Q_{yi} (y_{sp,i} - y_i) + \ldots \\ & + (u_{i-1} - u_{r,i-1})^T Q_{ui} (u_{i-1} - u_{r,i-1}). \end{aligned} \tag{5}$$

Here $y_{sp}$ is the setpoint and $u_r$ is a reference input compatible with this setpoint. Note that $\chi(x_{i-1}, u_{i-1})$ is obtained through the evaluation of (1, 2) inside the sampling interval $t \in [t_{i-1}, t_i]$ only. To simplify the notation further, we define the concatenated vectors $U$, $X$ and $Y$ containing all values of the corresponding variables inside a predictive horizon:

$$\begin{aligned} U &\equiv \left[ u_0^T \ u_1^T \ \cdots \ u_{m-1}^T \right] \in \mathsf{R}^{\mathsf{n_s m}}, \\ X &\equiv \left[ x_1^T \ x_2^T \ \cdots \ x_p^T \right]^T \in \mathsf{R}^{\mathsf{n_s p}}, \\ Y &\equiv \left[ y_1^T \ y_2^T \ \cdots \ y_p^T \right]^T \in \mathsf{R}^{\mathsf{n_o p}}. \end{aligned} \tag{6}$$

As a result, the NLP can be rewritten as:

$$
\begin{aligned}
&Min_{U,X,Y}\, J(Y,U) \\
&s.t.\ F(X,U)=0,\ \ Y-G(X)=0 \\
&U_l \le U \le U_u,\ \ X_l \le X \le X_u,\ \ Y_l \le Y \le Y_u
\end{aligned} \tag{7}
$$

Note that this formulation is general enough to allow for the treatment of endpoint constraints or other devices to ensure stability of the nonlinear controller. It should be noted that (7) can now be solved using any NLP algorithm. In this paper we explore Newton-type approaches, as these methods can be adapted to exploit the structure of the NMPC problem. These approaches have the advantage of simplicity along with well-known convergence properties. Moreover, unlike general purpose NLP solvers, they can be easily tailored to different applications, with significant benefits in performance. As a result, we briefly summarize the Successive Quadratic Programming (SQP) algorithm for (7) in the next section.

### 2.1 Basic SQP Algorithm

Using the Successive Quadratic Programing (SQP) algorithm for problem (7), the following quadratic programming (QP) subproblem is created and solved at iteration $k$, and the solution, $d$, is used as a search direction to converge to the solution of (7):

$$
\begin{aligned}
&Min_d\ \nabla_Y J(Y_k,U_k)^T d_Y + \nabla_U J(Y_k,U_k)^T d_U + 1/2 d^T B_k d \\
&s.t \\
&F(X_k,U_k) + \nabla_X F(X_k,U_k)^T d_X + \nabla_U F(X_k,U_k)^T d_U = 0 \\
&Y_k + d_Y - G(X_k) - \nabla_X G(X_k)^T d_X = 0 \\
&U_l \le U_k + d_U \le U_u, X_l \le X_k + d_X \le X_u \\
&Y_l \le Y_k + d_Y \le Y_u,
\end{aligned} \tag{8}
$$

where $d^T = [d_U^T \quad d_X^T \quad d_Y^T]$ and $B_k$ is the Hessian of the Lagrange function, $\nabla^2 L(U,X,Y,\lambda,\gamma) = \nabla^2 J + \Sigma_i \nabla^2 F_i \lambda_i - \Sigma_j \nabla^2 G_j \gamma_j$ or its approximation. As with any Newton-type method, asymptotic convergence properties for SQP are also well-known (see e.g. Fletcher, 1987). If $B_k$ is asymptotically equivalent to the actual Hessian at the optimum, $\nabla^2 L^*$ then the convergence rate is Q-quadratic, as is expected with any Newton method. However, $\nabla^2 L^*$ is rarely positive definite and this can lead to numerical difficulties in solving the QP, (8). Using a positive definite quasi-Newton approximation for $B_k$ (common in many generic SQP solvers), leads at best to only (one- or two-step) Q-superlinear convergence. Finally, a less faithful approximation for $B_k$ leads to only Q-linear convergence.

Similarly, a number of reliable stabilization algorithms are available to promote convergence from poor starting points (Fletcher, 1987). First, it is well known that if the $B_k$ matrix is positive definite in the reduced space of the active constraints, then $d$, the solution to (8 ), forms a strong descent direction for an exact penalty (or merit) function, the minimum of which corresponds to a solution of (7). Thus a stepsize along $d$ can be found which is bounded away from zero and the convergence argument then proceeds in a straightforward way.

However, these properties do not guarantee convergence from poor starting points. First, if (7) is nonconvex, a solution to (8) cannot be guaranteed, as the region of linearized constraints may not be feasible. In this case, a trick that allows the SQP algorithm to continue is to introduce an additional scalar variable $\xi$ and solve, for instance:

$$
\begin{aligned}
&Min_{U,X,Y} \quad J(Y,U) + M\xi \\
&s.t. \quad F(X,U)(1-\xi) = 0 \\
&\quad (Y - G(X))(1-\xi) = 0 \\
&\quad U_l \le U \le U_u, \quad X_l \le X \le X_u \\
&\quad Y_l \le Y \le Y_u, \quad 0 \le \xi \le 1,
\end{aligned} \tag{9}
$$

where $M$ is chosen suitably large to keep $\xi$ as small as possible. The modified problem has a feasible point with the variables between their bounds and with $\xi = 1$. Under the above conditions, solution of (9) either satisfies the constraints of (7) or $\xi^* = 1$, and the stationary point is infeasible.

Second, the SQP algorithm attracts to (first order) Kuhn-Tucker points that may not be a local minimum. In fact, simply tracking Kuhn Tucker points may even lead to a local maximum (Mayne,1996). To ensure optimum solutions, however, one can examine the second order conditions at the converged point and follow directions of negative curvature if the point is not optimal. These directions ensure an immediate descent away from the converged point. Alternately, SQP methods have been developed that, under mild conditions, will attract to local minima. These usually require exact second derivatives as well as trust region strategies (Byrd et al., 1996).

### 2.2 Exploitation of Structure for NMPC

For NMPC, we note that (7) has an exploitable structure that can accelerate the SQP algorithm. For instance, if we assume a regulatory problem where the setpoint is interior to the bound constraints, then the Kuhn Tucker conditions for this problem are:

$$\nabla_Y J + \gamma = 0\ , \tag{10}$$

$$\nabla_U J + \nabla_U F \lambda = 0\ , \tag{11}$$

$$\nabla_X F \lambda - \nabla_X G \gamma = 0\ . \tag{12}$$

From the first equation we can rewrite the relations as:

$$\nabla_U J + \nabla_U F \lambda = 0\ , \tag{13}$$

$$\nabla_X G \nabla_Y J + \nabla_X F \lambda = 0\ , \tag{14}$$

and since the evolution of the states is well defined, $\nabla_X F$ is square and nonsingular and $\lambda =$

$-(\nabla_X F)^{-1}\,\nabla_X G\,\nabla_Y J$. Consequently, if the predicted outputs remain close to their setpoints, $\nabla_Y J = Q_Y(Y - Y_{sp}) \approx 0$ and $(\gamma, \lambda) \approx 0$ as well. As a result, we can simplify the evaluation of the $B_k$ matrix in (8) since $\nabla^2 L(U, X, Y, \lambda, \gamma) \approx \nabla^2 J = Q$ where

$$Q = \begin{bmatrix} Q_U & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & Q_Y \end{bmatrix}, \tag{15}$$

$Q$ is positive definite in the reduced space of the linearized equality constraints and setting $B_k = Q$ therefore leads to a convex QP with a unique solution.

This approach, termed the Gauss-Newton property, is well known for unconstrained least squares problems and has been demonstrated very successfully for constrained least squares problems (Bock, 1983). In the case of zero residuals and nonlinear constraints, this simplification leads to a quadratic convergence rate. This rate becomes linear if projected contributions of the nonlinear constraints are nonzero, but in practice convergence is still quite good for problems with small residuals. Moreover, for large residuals variations of this approach also include hybrid strategies (Fletcher, 1987; Albuquerque and Biegler, 1995, 1995) which adapt and switch between quasi-Newton methods and the Gauss-Newton approximation for $Q$.

Solution of (7) can be simplified by considering the forward structure of the equality constraints. Given the availability of efficient solvers for initial value DAEs (e.g., DASSL, DDASAC), it is straightforward to combine general purpose NLP software with these solvers to develop NMPC controllers. More efficient approaches can also be developed by considering the properties of the previous section and noting some parallels with linear MPC controllers.

### 2.3 Sequential Approach

Because of the relation to convergence properties of SQP, the Gauss-Newton approximation leads to much faster NMPC algorithms than with general purpose NLP solvers. Moreover, for problems that converge to the setpoint (zero residuals), this leads to a Q-quadratic convergence rate. To conclude this section, we consider a sequential, reduced space approach where the elimination of linearized equality constraints in (8) leads to a smaller inequality constrained QP entirely in $d_U$:

$$\begin{aligned} Min_{d_U}\, & a_k^T d_U + 1/2 d_U^T (S^T Q_Y S + Q_U) d_U \\ s.t. & U_l \le U_k + d_U \le U_u \\ & X_l \le X_k + r_k + R d_U \le X_u \\ & Y_l \le Y_k + s_k + S d_U \le Y_u \end{aligned} \tag{16}$$

Here $a_k = (\nabla_U J(Y_k, U_k) + S^T(\nabla_Y J(Y_k, U_k) + Q_Y s_k))$, $r_k = -\nabla_X F^{-T} F_k$, $R = -\nabla_X F^{-T} \nabla_U F^T$, $s_k = (G_k - Y_k) + \nabla_X G^T r_k$, and $S = \nabla_X G^T R$. These terms are derived from a forward elimination of the state equations which corresponds to an initial value problem (IVP) decomposition. This decomposition leads to the Newton-type control algorithm for nonlinear MPC described in our previous work (Oliveira and Beigler, 1985) and is a nonlinear analog to the QDMC algorithm for linear models. Because the controller has a starting point based on the solution from the last sampling time, it usually requires very few iterations of (16) to converge (7). In fact, a single solution of (16) is often sufficient to obtain good performance of the nonlinear controller (Li and Biegler, 1989; Oliveira and Biegler, 1985).

## 3. A Simultaneous Approach for NMPC

While solution of (16) is efficient for nonlinear systems that are stable in the large, severe problems can be encountered if (16) is solved for locally unstable systems. This instability is encountered in two ways. First, integration of a DAE model (1,2) at an unstable point leads to unbounded profiles and immediate failure of the NLP algorithm. In certain process systems (e.g., closed loop systems with poorly tuned controllers, exothermic reactors) it may be difficult to find inputs that give bounded profiles except very close to the optimal solution (Bock, 1983; Biegler, 1998) and here the sequential approach is unsuitable for dynamic optimization.

The second problem occurs even if we obtain a bounded solution. This occurs in the forward decomposition of multiple shooting problems applied to (7). Here even though functions and gradients can be evaluated for the NLP solver, the influence of unstable modes leads to $R$ and $S$ matrices that are very poorly conditioned. For linear problems, this instability is treated by matrix factorizations that cancel the unstable modes (see Rao et al., 1998). On the other hand, nonlinear problems must be treated through boundary value problem (BVP) formulations, and these require us to return to (8). In this larger system (8), the state profiles remain bounded for integration of the predictive horizon and trust region methods can also be applied to promote numerical stability. Here well conditioned formulations are ensured when the unstable forward modes are pinned down by final conditions. This complements the well-shown endpoint constraint formulations of (Keerthi and Gilbert, 1988; Rawlings and Muske, 1993) where the constraint $x_{k+p} = 0$ is imposed in (7); it also applies to dual mode controllers (Michalska and Mayne,1993) which confine the endpoint to a region around the setpoint.

"Pinning down" the unstable modes is characterized by the concept of dichotomy, where stable modes either in the forward or reverse direction are identified with their respective boundary conditions. In fact, well-conditioned systems and dichotomous systems can be shown to be equivalent (deHoog and Mattheij, 1989).

Consequently, by ensuring well-conditioned systems, through careful choice of the pivot sequence and scaling, one can also ensure a dichotomous problem formulation. We can apply this strategy to (7) through linearization of the (now discretized) DAE model, $c(w_k) + \nabla c(w_k)^T d = 0$. By choosing independent (inputs) and dependent (state) $w$ variables and partitioning $\nabla c(w_k)^T = [N|C]$ we effectively define the boundary conditions for this problem and solve for the state variables by using a well-conditioned $C$ matrix as the state Jacobian. Necessary conditions for finding $C$ include determination of a stable pivot sequence and rescaling the system, and these can be done with standard linear algebra software. On problems with few degrees of freedom (e.g., NMPC problems with short input horizons), the partitioning can be combined with a reduced space SQP approach (with subproblems similar to (16) and a reliable and efficient NLP strategy results. Further details of this approach are given in (Cervantes and Biegler, 1998). Moreover, this study also considers the dynamic optimization of unstable reactors and reactive separation systems.

In addition to exploiting the convergence properties of SQP and the conditioning of the QP subproblem, we also need to consider efficient strategies for the solution of (8). For structured QPs there is a tremendous difference in the performance of general purpose QP strategies vs. tailored methods. If we now consider the Kuhn-Tucker matrix of (8), without bounds, we have a system of order $O((n_s + n_i + n_o)p)$. With a general purpose solver the computational effort is $O(\{(n_s + n_i + n_o)p\}^3)$. Considering a smaller input horizon and using either (16) or (8), where a decomposition is applied to solve the problem in the space of $d_U$, leads to computational effort that is proportional to $O\left((n_i m)^3 + [(n_s + n_o)\, p]^\tau\right)$, where $\tau = 3$ unless a sparse factorization is used for the linearized state equations. However, even this performance can be unattractive for large systems.

### 3.1 Matrix Structure of NMPC

To tailor the solution of these QPs we consider the structure of (7) in greater detail. Using equation (4) to define $J$ we can reorder the Kuhn-Tucker matrix from (8) to form an almost block diagonal (ABD) system:

$$\begin{bmatrix}
Q_1 & A_1^T & C_1^T & 0 & & & & & & \\
A_1 & 0 & 0 & \bar{A}_1 & & & & & & \\
C_1 & 0 & 0 & 0 & & & & & & \\
0 & \bar{A}_1^T & 0 & Q_2 & A_2^T & C_2^T & 0 & & & \\
 & & & A_2 & 0 & 0 & \bar{A}_2 & & & \\
 & & & C_2 & 0 & 0 & 0 & & & \\
 & & & 0 & \bar{A}_2^T & 0 & \ddots & & & \\
 & & & & & & & Q_n & A_n^T & C_n^T \\
 & & & & & & & A_n & 0 & 0 \\
 & & & & & & & C_n & 0 & 0
\end{bmatrix}, \qquad (17)$$

where the matrices are defined as $\Phi_i = \frac{\partial \chi(x_i, u_i)}{\partial x_i}$, $\Gamma_i = \frac{\partial \chi(x_i, u_i)}{\partial u_i}$, $D_i = \frac{\partial g(x)}{\partial x}$, $A_i = [\Gamma_i \ \Phi_i \ 0]$, $\bar{A}_i = [0 \ -I \ 0]$ $C_i = [0 \ -D_i \ I]$ and

$$Q_i = \begin{bmatrix} Q_{ui} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & Q_{yi} \end{bmatrix}. \qquad (18)$$

This ABD system can be decomposed efficiently using a number of methods, including:

- A symmetric sparse solver such as MA27 or MA47 (Harwell Subroutine Library). This approach scales with a low power of the problem size (usually between one and two) and leads to quite stable decompositions.
- Nonsymmetric sparse solvers such as MA28 or MA48, can also be used. These routines require more analysis of the pivot sequence but this is an offline cost. In fact, the online factorization times with these solvers are usually less than with symmetric sparse solvers.
- Dunn and Bertsekas, 1989, represented the multipliers and the changes in the control variables as affine functions of the changes in the state variables. For stable systems, this Riccati-like decomposition leads to a backward and forward recursion for the multiplier and state variables, respectively.
- The SOLVEBLOK approach (deBoor and Weiss, 1980) can be applied to (17) if the system is unstable. However, neither decomposition exploits sparsity within each block.

Given the complexity arguments for the above decomposition strategies, NMPC strategies can be solved with much faster performance than by using general purpose NLP solvers or even the modified linear MPC solver (16). On parameter and state estimation problems with the same ABD structure, performance improvements over general purpose NLP and QP solvers greater than three orders of magnitude were obtained (Albuquerque and Biegler, 1997). Clearly the exploitation of this problem structure is often the key to getting these kinds of improvements for NMPC. However, to deal with inequalities on the inputs and outputs, we also need to develop constraint handling strategies that preserve the structure of (17), as discussed next.

### 3.2 Constraint Handling for NMPC

Inequality constraints for (8) can be treated either by active set strategies or interior point (IP) methods. A detailed discussion on the pros and cons of both methods for model predictive control is given by (Wright, 1997). Active set strategies are most commonly used in general purpose QP solvers where the active constraint set is modified through rank one updates of the Kuhn-Tucker matrix. One disadvantage of active set strategies is that they add a combinatorial element to the identification of the active set and the solution of the

QP, and this leads to a large increase in computational effort with problem size. Performance characteristics of the interior point method, on the other hand, seem to be insensitive to problem size. For instance, with a small NMPC example (Albuquerque et al., 1999) with 1000 constraints MINOS and QPSOL performnace increases polynomially to over 1500 iterations while the IP method remains at a constant 6 iterations.

To illustrate the application of IP methods, we reconsider the QP subproblem (8) where we combine all of the linearized equality constraints into $w_k + W_k d = 0$ and the variable bounds into inequality constraints: $h_k + H_k d \leq 0$. Introducing slack variables $s$ into these constraints (and suppressing the subscript $k$) leads to a quadratic program written as:

$$\begin{aligned} min & c^T d + \frac{1}{2} d^T Q d \\ s.t. & w + Wd = 0 \\ & h + Hd + s = 0, \quad s \geq 0, \end{aligned} \tag{19}$$

where $c^T = [\nabla_U J(Y_k, U_k)^T,\ 0,\ \nabla_Y J(Y_k, U_k)^T]$ and

$$W = \begin{bmatrix} \nabla_U F(X_k, U_k)^T & 0 & \nabla_Y F(X_k, U_k)^T \\ 0 & \nabla_X G(X_k) & -I \end{bmatrix}$$

The optimality conditions of (19) are given by (20)-(23).

$$Qd + W^T \lambda + H^T \nu + c = 0\ , \tag{20}$$

$$w + Wd = 0\ , \tag{21}$$

$$h + Hd + s = 0\ , \tag{22}$$

$$SVe = 0, \quad (s, \nu) \geq 0, \tag{23}$$

with $e^T = [1, 1, \ldots, 1]$, $\lambda$ and $\nu$ as the multipliers of the equalities and inequalities respectively. Also, $S = diag\{s\}$ and $V = diag\{\nu\}$. Note that this system is nonlinear because of the complementarity conditions given in the last equation.

To solve this system reliably, a central path is introduced, which keeps the iterates biased towards the interior of the nonnegative region $(s, \nu) > 0$, and also forces all the complementarity pairs to converge at a similar rate. Here the complementarity gap $\mu$ is a measure of the average value of the products $s_i \nu_i$ and is defined as:

$$\mu = s^T \nu / n. \tag{24}$$

To keep the iterates in the interior, (23) is replaced with (25) where $0 \leq \sigma \leq 1$ is a parameter.

$$SVe = \sigma \mu e. \tag{25}$$

Setting $\sigma = 1$ corresponds to a pure centering direction, i.e. moving all the pairwise products to the current average value $\mu$, while $\sigma = 0$ gives a standard Newton step (20)-(23). Various path following methods differ in the way the search direction is modified and $\sigma$ is chosen. An excellent survey and analysis of these methods is given in (Wright, 1997).

The most widely used approach for selecting $\sigma$ to solve IP equations is the Mehrotra predictor-corrector algorithm (Mehrotra, 1993). Each Mehrotra iterate solves an affine scaling predictor step (26) and a corrector step (27), where $r_1, r_2, r_3$ are the residuals of (20)-(22) at the current point. The predictor step helps to reduce the complementarity gap while centering is accomplished through the corrector step, which also incorporates a second order correction term in (27)

$$\begin{bmatrix} Q & W^T & H^T & 0 \\ W & 0 & 0 & 0 \\ H & 0 & 0 & I \\ 0 & 0 & S & V \end{bmatrix} \begin{bmatrix} d^p \\ \Delta\lambda^p \\ \Delta\nu^p \\ \Delta s^p \end{bmatrix} = - \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ SVe \end{bmatrix}, \tag{26}$$

$$\begin{bmatrix} Q & W^T & H^T & 0 \\ W & 0 & 0 & 0 \\ H & 0 & 0 & I \\ 0 & 0 & S & V \end{bmatrix} \begin{bmatrix} d^c \\ \Delta\lambda^c \\ \Delta\nu^c \\ \Delta s^c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \sigma\mu e - \Delta S_p \Delta V_p e \end{bmatrix}. \tag{27}$$

For the solution of linear programs, IP methods have been developed into powerful, commercial grade solvers. These have also been extended to the solution of quadratic programs and convex NLPs in general. For example, an efficient strategy has been developed that applies the Mehrotra predictor-corrector to linear MPC problems (Rao et al., 1998). Here a Riccati transformation (with a linear relation between the $d$ and $\lambda$) was used to decompose the matrices in (26)-(27).

While the number of IP iterations is not a function of problem size, the cost per IP iteration is generally much larger than simple rank one updates to active sets; as a result IP methods are not a universal successor to active set methods. However, (26)-(27) can be rearranged along the lines of the ABD structure in (17) so for NMPC problems, where a large number of active constraints are expected, significant benefits can be obtained with IP approaches.

This approach was illustrated in (Albunquerque et al., 1999) using the Tennessee Eastman Reactor problem. This control example consists of a two phase, jacketed CSTR, with the reaction dynamics occurring in the vapor phase. The DAE model given in (Albunquerque et al., 1999) consists of 8 differential and 28 algebraic equations and two control profiles. Because this problem is open loop unstable, it must be handled with a simultaneous formulation. The nonlinear system was solved with a full space SQP method with the QP subproblems solved using an IP method that exploits the ABD structure. Further information on this problem can be found in (Albunquerque et al., 1999). Here, sparse symmetric solvers MA27 and MA47, and the nonsymmetric solver MA28 from Harwell were applied. In this initial solution of the nonlinear optimal control problem the initial point was chosen close to the steady state values (as is typical with NMPC applications). As a result relatively few SQP iterations were required. Performance using a Mehrotra predictor-corrector implementation is given in (Albunquerque et al., 1999).

About 10 IP iterations were required to solve each QP problem (8) and NLPs with up to 35,000 variables were solved in under 11 CPU minutes (DEC AlphaStation).

Current research deals with improving the solver efficiency for NMPC problems. For instance, Bock et al. introduce a novel strategy for re-using the ABD structure as the time horizon is shifted from one sampling point to the next. As much of the matrx and its factorization remains unchanged, this re-initialization leads to a significant improvement in efficiency. In addition, the interior point strategy can be improved by "breaking the QP loop" and applying the interior point formulation at the NLP level. This approach currently represents an active research area with the potential for great computational savings. For example, in preliminary work Cervantes et al. (1999) solved a dynamic optimization problem with 426 DAEs in under 10 CPU minutes (DEC AlphaStation). Moreover, NMPC algorithms can be made even more efficient through the substitution of simpler dynamic models. This is discussed in the next section.

## 4. NMPC With DABNET Models

We now consider the use of nonlinear discrete DABNet (De-coupled A-B Neural network, Sentoni et al., 1998) models in a Nonlinear Model Predictive Control (MPC) scheme. Using NMPC often requires a discretization in time of the differential algebraic equations describing the process to be controlled. This leads to a set of difference equations that form constraints in the receding horizon control problem. The resulting nonlinear programming problems can be exploited using Newton type solvers which take advantage of an almost block diagonal linear system and a number of specializations can be made to the optimization problem, including simplification of the Hessian matrix and application of interior point methods to deal with input and output variable bounds (Albunquerque et al., 1999). On the other hand, even with these performance improvements, there will always be more severe limits due to the computational time for on-line application. This leads to a trade-off between the time required for a large, detailed model and the need for small sampling times for control performance.

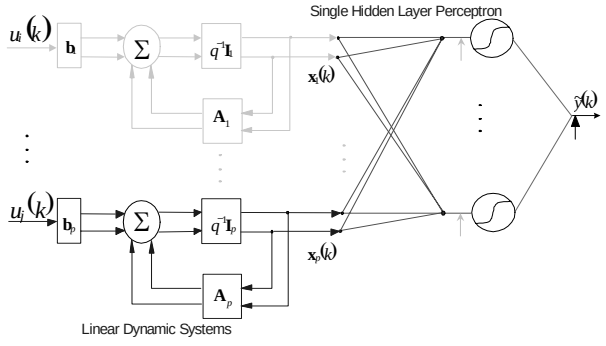



Fig. 1. DABNet Structure

To deal with this trade-offff, we explore a hybrid approach where first principle dynamic optimizations are conducted off-line to determine desirable output, state and input trajectories. All of these reference trajectories will then be used online with an approximated, empirical model that will cope with the disturbances and the plant/offline model mismatch. We then apply DABNet models to an MPC strategy. These models are composed of a linear dynamic system whose states, de-coupled by inputs, are mapped by a single hidden layer Perceptron. This is illustrated by Figure (1) and is represented by the following MISO state space model:

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k) \\ \tilde{y}(k) &= \mathbf{f}_{NN}(\mathbf{x}(k)) + v(t), \end{aligned} \tag{28}$$

where $\tilde{y}(k)$ is the DABNet output, $v(t)$ is the measured disturbance and $\mathbf{f}_{NN}$ a nonlinear functions that represents the neural network. In previous work (Sentoni et al., 1998), it was shown that the DABNet structure is able to approximate any nonlinear, causal, discrete time-invariant, multi-input, single-output process with fading memory. This description fits a wide variety of typical processes for which the DABNet models are well suited to be embedded into a Model Predictive Control Strategy. Because of the discrete nature of the models, the MPC problem can be reformulated as the following (NLP):

$$\begin{aligned} &Min_{\mathbf{u}(k),\ldots,\mathbf{u}(k+P-1)} \; F = F_1 \\ &s.t. \\ &\quad v(i) = \mathbf{y}(k) - \tilde{\mathbf{y}}(k), \quad i : k, \ldots, k+p \end{aligned} \tag{29}$$

where:

- $F_1 = \sum_{i=k+1}^{k+p} \|\tilde{\mathbf{y}}(i) + v(i) - \mathbf{y}^o(i)\|_q^2$
  $\mathbf{y}^o(i) = \mathbf{y}^o(t)$ for $t = t_s$ (being $t_s$ the sample time),
  $v(i)$ : is a vector that accounts for model-plant mismatch,
  $p$ : is the Prediction Horizon.

In practice, constraints must be placed on the input control and output variables. There are usually hard constraints on the control inputs so that they are forced to lie within specified operating limits. Due to the possibility of infeasible solutions, output constraints are treated as soft constraints. These can be handled using the Exact Penalty treatment found in (Oliveira and Biegler, 1995). Here if the penalty parameter is sufficiently large, this treatment guarantees that the constraints are satisfied provided there is a feasible solution. Moreover, large control steps are usually undesirable and therefore a move suppression capability is an important component of the formulation:

$$
\begin{aligned}
&Min_{\mathsf{u}(k),\ldots,\mathsf{u}(k+P-1)}\ F = F_1 + F_2 \\
&s.t. \\
&\qquad \upsilon(i) = \mathsf{y}(k) - \widetilde{\mathsf{y}}(k), \quad i : k, ..., k+p \\
&\qquad \mathsf{u}(i+1) = \mathsf{u}(i), \qquad i > k+l \\
&\qquad \mathsf{u}^L \leq \mathsf{u}(k) \leq \mathsf{u}^U
\end{aligned} \tag{30}
$$

where:

- $F_2 = \sum_{i=k}^{k+l} \|\mathsf{u}(i) - \mathsf{u}(i-1)\|_r^2$
  $\mathsf{u}^L \leq \mathsf{u}(k) \leq \mathsf{u}^U$ : lower and upper bounds for the control variable,
  $l$ : is the Control Horizon.

In addition the control variables themselves may also have ideal resting points based on similar considerations:

$$
\begin{aligned}
&Min_{\mathsf{u}(k),\ldots,\mathsf{u}(k+P-1)}\ F = F_1 + F_2 + F_3 \\
&s.t. \\
&\qquad \upsilon(i) = \mathsf{y}(k) - \widetilde{\mathsf{y}}(k), \quad i : k, ..., k+p \\
&\qquad \mathsf{u}(i+1) = \mathsf{u}(i), \qquad i > k+l \\
&\qquad \mathsf{u}^L \leq \mathsf{u}(k) \leq \mathsf{u}^U
\end{aligned} \tag{31}
$$

where:

$$F_3 = \sum_{i=k}^{k+p} \|\mathsf{u}(i) - \mathsf{u}^o(i)\|_s^2$$

This NMPC approach ha been used to solve a wide variety of test problems (and has also been incorporated into a commercial product by Aspen Technology). In particular, this approach has been used for to reduce $NO_x$ and CO emissions of pulverized coal fired power plant.. Here the DABNet model was developed from plant tests and identification, as first principle models were very difficult to obtain for this combustion process. The results of this study show a 15-25% reduction in these emissions and leads to an increase in boiler efficiency for the power plant (Hong et al., 1999).

The DABNet structure can also be incorporated to run with first principle models. In addition, off-line optimization results using detailed first principle models can be used very effectively for NMPC algorithms that incorporate DABNet models. This is illustrated next with a nonlinear example.

## 5. Optimal Start-Up of a Polymerization Reactor

Here we consider the optimal start-up of a polymerization reactor. The process consists of a polymerization solution of methyl-methacrylate (MMA), initiated by azo-bis-isobutotyronitrile (AIBN), in which toluene is used as the solvent (Soroush and Kravaris, 1992). To begin, a first principle model was optimized using and SQP algorithm to solve (7). Then the identification of nonlinear DABNet MISO models was performed for the conversion factor $x_p$ and the temperature $T$ in the reactor, using the inputs and outputs of first principle model as a test case. The heater power $P$ and the initiator flow rate stream $F_i$ are the inputs for two DABNet models built to reproduce $x_p$ and $T$.

These DABNet models are then considered for three situations. Based on the value of the poles, the prediction horizon $p$ was chosen to be $p = 50$ following the guidelines given in (Sentoni et al., 1996). The control horizon $l$ was chosen to be $l = 13$. Based on these values, we consider the following scenarios:

**Equally weighted outputs and no suppression factor.** In this case the problem is stated as in equation (29) with $p = 50$, $l = 13$, and equally weighted outputs i.e. $q = diag\{1, 1\}$ in $\|.\|_q$, and $r = diag\{0, 0\}$ in $\|.\|_r$. The results are given in Figure 8, with an obvious oscillatory nature for the solution. This oscillatory effect can be smoothed with the incorporation of suppression factor.

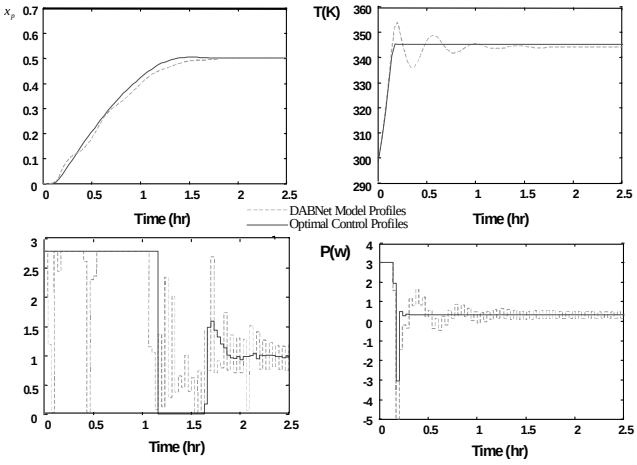



Fig. 2. DABNet Predictive Control: no weights

**Input Suppression factor applied.** In this case we have equally weighted outputs i.e. $q = diag\{1, 1\}$ in $\|.\|_q$, and and $r = diag\{0.05, 0.5\}$ in $\|.\|_r$. The results can be appreciated in Figure 9. In this case the solution is much closer to the optimal profiles than before.

**Suppression factor applied to the inputs.** This case shows the addition of the inputs to the minimized objective function, which themselves have ideal values from the first principle optimization. This is equivalent to minimizing on-line the evolution of the system with respect to $(\mathsf{y}^o(i), \mathsf{u}^o(i))$, the optimal trajectories calculated off-line. The objective function minimized is (31) with $p = 50$, $l = 13$, and $q = diag\{1, 1\}$ in $\|.\|_q$, $r = diag\{0.05, 0.5\}$ in $\|.\|_r$ and $s = diag\{0.1, 0.1\}$ in $\|.\|_s$. The results can be appreciated in Figure 10, which shows them to be the closest solutions to the optimal offline solution. It is interesting to say just a few words about speed of calculation. One way to do that is to compare a model predictive control approach solved by iteratively using orthogonal collocation, with a DABNet model predictive control. On the one hand, the optimal profiles of Figure 4 and Figure 5 demand 15 minutes to be solved (DEC Alpha 400). While the

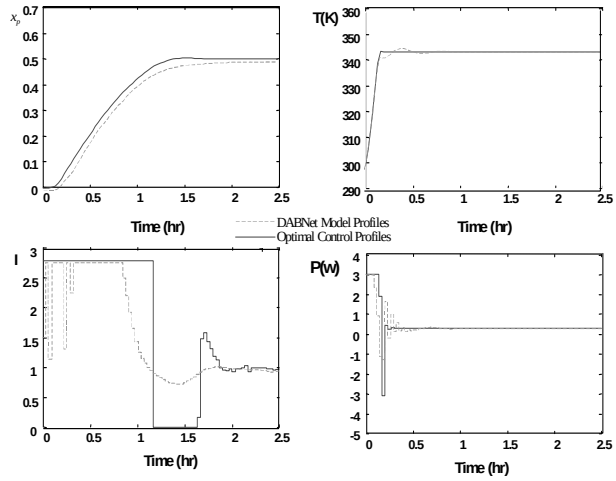



Fig. 3. DABNet Predictive Control: weights in the Outputs and in the Difference of Inputs

offline solution does not take full advantage of the refinements developed in Section 3, this approach is too expensive for the short sampling time($100s$) in this example to cope with model mismatch and disturbances. On the other hand, the optimization problem of Equation (31) was easily solved on-line. This shows a possible way of using calculations performed off-line (by calculating the optimal path) with calculations performed on-line (DABNet model predictive control) to cope with model mismatch and disturbances.

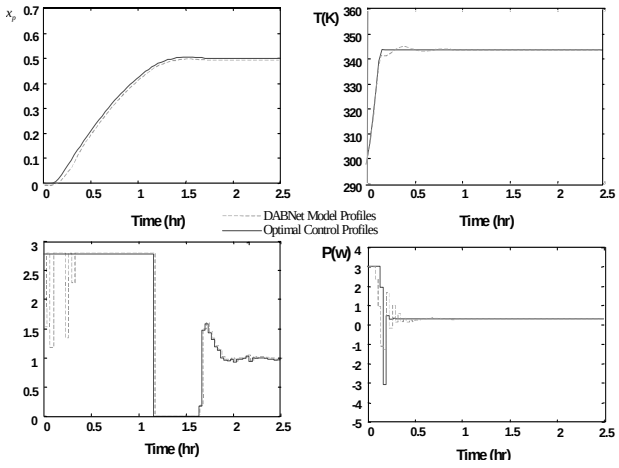



Fig. 4. DABNet Predictive Control: weights in the Inputs and the Outputs

## 6. Conclusions and Future Work

Optimization formulations have always been the key to describing complex process control problems. In previous years, however, difficulties and inefficiency in solving large nonlinear programs have caused control researchers to shy away from direct solution of these optimization formulations. Instead, clever problem reformulations have been proposed in the hope of avoiding the optimization problem altogether.

This paper reviews a number of recent developments and properties for the solution of NLPs for dynamic optimization problems. Included in this summary is an informal description of local and global convergence properties as well as the Gauss-Newton property, which can be applied to many NMPC problems. These lead to fast, large-scale optimization algorithms tailored to the NMPC problem. Moreover, a number of on-line applications have already been considered in industry that make use of these NLP formulations.

Dealing with large, complex NMPC problems also involves the exploitation of structure for nonlinear dynamic problems. Here the focus is on simultaneous problem formulations, as they allow reliable consideration of locally unstable and ill-conditioned systems. The NMPC problem with many degrees of freedom and application of the Gauss Newton properties can be solved efficiently with a full space SQP algorithm and a variety of efficient decomposition schemes can be applied to the Kuhn-Tucker matrix. Moreover, interior point techniques can be applied to handle bound constraints in the QP subproblem and still preserve the original structure of the Kuhn-Tucker matrix.

Finally, we consider the use of nonlinear discrete DABNet models embedded in a Nonlinear Model Predictive Control (NMPC) scheme for process control. A hybrid approach, where first principle optimizations are conducted off-line to determine desirable output, state and input trajectories was explored. Using this approach, the solution of the MPC problem with DABNet model was capable to reproduce the optimal control trajectories quite successfully, and leads to a very efficient on-line NMPC strategy.

Future research will lead to many more applications of dynamic optimization for process control. With advances in modeling and problem formulations, solution algorithms at various levels, better convergence properties and computer software and hardware, it is clear that much larger and more complex nonlinear systems will be considered.

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